



ANSWER SHEET MAPPER

One upload. Every answer, found and mapped.

Question extraction · handwritten answer extraction · label-based mapping · exact-region highlighting · **AI grading**

THE GOAL

A teacher should upload a paper and instantly know three things.



Which question

was answered — and which weren't, shown as its own visible state, not silently missing.



Where the answer is

highlighted as the exact region of the handwritten page, not the whole sheet.



How it was graded

a score, a verdict, and one line of feedback per question — optional, and honest about what it is.

APPROACH

One pipeline, six stages



Rasterize

PDF or images → PNG pages
(mupdf.js, WASM)



Extract questions

Printed order preserved,
sub-parts split



Extract answers

Text + bounding box
per handwritten region



Map by label

Normalized label match,
not position



Grade

Optional: score, verdict,
feedback per question



Highlight

Click a question →
exact region lights up

Pages within each document extract concurrently, not one-at-a-time — required once a real multi-page upload started exceeding a serverless time limit.

DEMO

Two files in, everything else automatic

A

Answer Sheet Mapper

Upload **question paper & answer sheet**

Extract questions, map handwritten answers, and highlight exactly where each one was written.

Question paper

PDF or one/more images

Answer sheet

PDF or one/more images

☒ Grade the answers (score, correct/incorrect, AI feedback)

Process answer sheet

- PDF or images — either works, for both files
- Live step-by-step progress while it runs, not a bare spinner
- Grading is opt-in, checked by default

DEMO

Click a question. See exactly where it was answered.

11. In regression analysis, the output (dependent variable) is typically: A. A ... **2/2**
12. The process of identifying the most relevant variables to use in a mode... **2/2**
13. Which technique is used to predict a continuous numerical value? A. CL... **2/2**
14. Analysis that involves data points collected or recorded at specific tim... **2/2**
15. Which of the following is actually a classification algorithm despite its ... **0/2**
16. Hyperparameters are best described as: A. Data labels B. Input feature... **2/2**
17. (True/False): Logistic regression is used for predicting continuous outc... **0/2**
18. In a confusion matrix, where are the "Correct Predictions" locate... **No answer**
19. What is the formula for Precision? A. $TP / (TP + FN)$ B. $TP / (TP + ...$ **No answer**
20. In Support Vector Machines (SVM), the data points closest to the hyper... **0/2**
21. Define Supervised Learning. **2/3**
- "Supervised learning is the modern language of algorithm model that already know output & input but, giving to input to machine and give predict the output and keep going until get high level of the output."*
- AI Feedback
The answer mentions learning from input-output pairs, but lacks clarity regarding training models on labeled datasets.
22. Evaluation Case Study: A model (M1) achieves 99% accuracy on t... **No answer**
23. What is Regression Analysis? **1/3**
24. List and elucidate two major challenges in Machine Learning. **No answer**
25. List any two major real-world applications of Data Science. **No answer**
26. Distinguish between Structured and Semi-structured data. **No answer**
27. Describe methods used to detect and handle outliers in a dataset. **0/4**

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①

I.

1. ①	11. B
2. ①	12. C
3. B	13. R
4. ①	14. C
5. ② B	15. A
6. ② B	16. C
7. A	17. A
8. A	18.
9. A	19.
10. B	20. C

II.

21. Supervised learning is the modern language of algorithm model that already know output & input but, giving to input to machine and give predict the output and keep going until get high level of the output.

22. Regression analysis is the process of machine learning that are trying to understand, learning, analyzing the relationship of data.

27. We can use outlier detection.

29. Data normalization is required in k-NN because it is process of preparation in which help organize the difficult to understand data into normal scale.

DEMO

Grading, summarized and per-question

Answer Sheet Mapper

Results

Start over

Total score	Correct	Partially correct	Incorrect	Answered / Total
20 / 49	8	3	11	22 / 35

1. Which of the following is an assumption of Linear Regression? A. High ...

2. In a perfectly fitted regression model, the sum of residuals (errors) shoul...

3. Which distance metric is specifically used for categorical variables or st...

4. Supervised learning can be split into which two main categories? A. Cla...

5. Which of the following is an ensemble learning method based on decisl...

6. The process of combining multiple classifiers to improve prediction (e.g. ...

7. In a Confusion Matrix, what is the value of the error for a perfect classifi...

8. Which visualization tool is most effective for identifying correlations bet...

9. In a scatter plot, individual data observations are represented as: A. Poin...

10. Which measure of central tendency is most sensitive to outliers? A. Me...

11. In regression analysis, the output (dependent variable) is typically: A. A...

12. The process of identifying the most relevant variables to use in a mode...

13. Which technique is used to predict a continuous numerical value? A. CL...

14. Analysis that involves data points collected or recorded at specific tim...

Answer Sheet

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Page 1Page 2

I.

1. D

2. D

3. B

4. D

5. D B

6. D B

7. A

8. A

9. A

10. B

11. B

12. C

13. B

14. C

15. A

16. C

17. A

18.

19.

20. C

II.

21. Supervised learning is the modern language of algorithm model that already know output & input but giving to input to machine and give predict the output and keep going until get high level of the output.

22. Regression analysis is the process of machine learning that are trying to understand, learning, analyzing the relation, value of data.

 Correct

full marks awarded

 Partially correct

some marks, feedback on what's missing

 Incorrect

zero marks, one line on why

Max marks per question are inferred from its own complexity — a one-line recall question and a multi-part "explain/compare" don't get the same denominator.

HANDLING EDGE CASES

Every case falls out of one design decision

Matching answers to questions by normalized label — not position — means none of these need special-case code.



Out of order

Answered 21, 23, 27, 29 — label lookup doesn't care about sequence.



Unanswered

Zero matches at that label → shown as its own grey state, not omitted.



No matching question

Collected separately as unmatched answers, never silently dropped.



Spans multiple pages

Same label across pages → flagged, both fragments listed with page numbers.



Split across a table

One answer in two columns, one label — fixed with a geometric row-adjacency check.



Sub-parts

"11(a)" and "11(b)" become separate entries, each keeping the parent number.

BUILT WITH VERIFICATION, NOT ASSUMPTION

Three real bugs, caught before they shipped



The boxes that looked wrong but weren't

Debug render made every box on a 20-row answer list look shifted up one row. Cropped the raw image at the claimed pixel coordinates before trusting that theory — the boxes were pixel-accurate. The label chip in my own visualization was the bug, not the extraction.



"11(a)" losing its parent number

Sub-part questions extracted as bare "(a)" / "(b)" — a student writes "11(a)", never bare "(a)", so matching would have silently failed for every sub-parted question. Caught on a synthetic test built specifically to exercise that one requirement line.



A student's table answer split in two

One question answered as a two-column table, number written once. The unlabeled half fell through as an orphan and graded as if the student never wrote it. Fixed with a geometric row-adjacency rule, not just a hopeful prompt tweak.

UNDER THE HOOD

Stack & reliability

STACK

Frontend + API

Next.js App Router, one repo (monorepo by construction)

AI model

Gemini gemini-3.6-flash — structured JSON output, temperature 0

PDF rendering

mupdf.js (WASM) — no native dependency, survives Vercel's serverless runtime

Deploy

Vercel, streamed NDJSON progress over a plain Response

RELIABILITY

Disk-cached Gemini calls

keyed on model+prompt+schema+image hash — re-runs during dev cost zero quota

Quota fallback

one automatic retry on a secondary model — never the default, since A/B testing found it measurably less accurate

Parallel page extraction

concurrent, not sequential — fixes a real timeout that hung the client with no error

Client stall watchdog

45s timeout on every stream read — a hang is now always a visible error

KNOWN LIMITATIONS

Said plainly, not discovered later



No answer key is uploaded, so grading is the model's own subject-matter judgment — not a comparison to ground truth.



Verified against one real scanned handwriting sample in depth, not a broad handwriting-style benchmark.



Label matching handles notation variants ("11(a)" vs "Ans. 11a"), not free-form paraphrasing of a label in words.



Free-tier Gemini quota and Vercel's request-body ceiling are real, external constraints — documented, with a fallback path for the first.



ANSWER SHEET MAPPER

Thank you.



Live app

answer-sheet-mapper.vercel.app



Repository

github.com/priyalrathore003/answer-sheet-mapper



Design doc

LLD.md — architecture, sequence, and the two core algorithms